

Sandwich panels for technical insulation

Technical datasheet

Revision: 1.0 / March 2026

Core material comparison: PUR · PIR · Rockwool (mineral wool)

Reference standards: EN 13165, EN 13162, FM 4880/4881, BS 476

Technical Metric	PUR (Polyurethane)	PIR (Polyisocyanurate)	RW (Rockwool / Mineral Wool)
Thermal Conductivity λ (W/m·K at 10°C)	0.022 – 0.026	0.020 – 0.024 (best)	0.035 – 0.045
Fire Reaction Class (EN 13501-1)	Euroclass E / F	Euroclass B-s3, d0 or B-s2,d0 (self-extinguishing)	Euroclass A1 / A2 (non-combustible)
Max Service Temperature (°C continuous)	~80°C	~120°C	~750°C (structural integrity > 1000°C)
Compressive Strength (kPa at 10% deflection)	100 – 140 kPa	120 – 150 kPa	60 – 100 kPa (lower)
Water Absorption (% by volume)	< 2.0%	< 1.5% (superior closed cell)	< 1.0%* *with hydrophobic agents; sensitive to cladding damage
Optimal Applications (recommended use cases)	Cold storage rooms, food processing facilities, temperature-controlled transport.	Industrial warehouses, high-efficiency commercial roofs, logistics centers, external façades.	Fire separation walls, marine insulation, high-heat hazard plants, acoustic enclosures.

Technical notes & practical insights

- Thermal conductivity and mechanical values may vary by manufacturer and panel thickness. **PIR** offers the best insulation performance with reduced thickness.
- Fire performance:** Rockwool is inert (class A1) – ideal for party walls and strict fire codes. PIR with Euroclass B-s3,d0 meets most industrial requirements where limited fire reaction is mandatory.
- Mechanical resistance:** PIR and PUR withstand point loads (maintenance, foot traffic). Rockwool has lower compressive strength; not suitable for frequent walking without extra protection.
- Aging & durability:** PIR retains its insulating properties longer than PUR, especially in elevated temperatures. Rockwool is indefinitely stable if kept dry.

- **Relative cost (indicative):** RW > PIR > PUR (rockwool is typically 20–40% more expensive than PIR, depending on thickness and density).
- EN 13165 (PUR/PIR)

EN 13162 (Mineral wool)

FM 4880/4881 (fire resistance for roofs)

BS 476-6/7 (flame propagation)

Technical reference document — Values represent typical European market data. For a specific project, always consult manufacturer-validated datasheets.